Mathematical overview

Independent nonnegative rectangle sides A in [a,b] and B in [c,d] give area in [ac,bd], tightly when each endpoint pair is allowed. A known sum links sides and changes the range: for whole sides at least 1 and A+B=6, labeled pairs (1,5),(2,4),(3,3),(4,2),(5,1) have areas 5,8,9,8,5. The safe independent enclosure [1,25] loses the shared-sum information; the tight enclosure is [5,9]. There are three shapes up to rotation, or five labeled side pairs. Intermediate real areas and whole-side enumeration are different questions.

Guaranteed ranges for the SAME unchanged hidden length intersect. Independent lengths do not. If exactly one card is false, its interval must actually fail for the chosen witness, while the other cards must hold; merely ignoring a card is not enough. The printed examples make every proposed false-card witness disjoint from that card, so their logic is unambiguous. No measurement distribution, averaging rule or confidence percentage is assumed.

For independently allowed endpoints A in [a,b], B in [c,d], distance |A-B| has minimum max(0,a-d,c-b) and maximum max(|a-d|,|b-c|). The minimum is zero if ranges overlap, including one shared endpoint. Maximum occurs at opposite endpoints. All edges are allowed; values can lie between ticks unless a task specifies whole units. A can be longer exactly when b>c, B can be longer exactly when d>a, and equality is possible exactly when ranges intersect. Strictly longer excludes a tie.

Grades 2-5 can tile rectangles, intersect card possibilities and slide endpoints. Ready older children explain why dependence changes extremes, diagnose false guarantees and design near/far ranges. Trials provide allowed witnesses or counterexamples; all-side enumeration, interval intersections and extremal arguments prove the tight bounds.

Readiness, kits, launch, and use

Unpiloted. Paper tiles and sleeve/strip handling have not been physically rehearsed. Page 1 needs whole lengths through 5, addition to 6, and counting up to 9 tiles; multiplication can grow from the model. Page 2 needs interpreting simultaneous statements about one length, retaining which cards are true, and testing false witnesses. Page 3 needs comparing two endpoints, including ties, and understanding separate choices. Reading and arithmetic can be adult-supported, but do not have an adult run the entire investigation. K-1 may explore tile rectangles or selected endpoint trials; no independent version of the logic-heavy work is forced.

Per pair: packet, twenty-five paper square tiles of side 20 mm, two sliding paper strips with opaque sleeves or cover cards, ruler, two endpoint markers, pencil and scrap paper. The page-1 working squares are exactly 20 mm, unlike its small non-task example. The page-2 ruler spacing is 45 points (15.9 mm) and page-3 spacing 39 points (13.8 mm); these are ideal unit diagrams, not millimetre-instrument uncertainty claims. Draw two equal physical unit rulers if using the same sleeves on both pages, or use endpoint markers instead. Strip range boundaries are stipulated guarantees. No calibrated error bound is inferred from a ruler's markings.

For eleven children in K–1 / 2–3 / 4–5, prepare five pair kits: 125 paper tiles, ten strips and sleeves/cover cards, five rulers, ten endpoint markers, five pencils, and three anchored adults. The third older child checks cards and rotates into the pair. Counting tiles on a separate table surface is fine; a printed grid is one working mat, not a requirement to maintain five overlapping drawings.

Launch, 4 minutes: let children build a rectangle with tiles. Follow the non-task 2-by-3 example to six square tiles; it has side sum 5, so it does not answer the sum-6 investigation. On a card visit, choose a non-task hidden length 2.5 and show that cards 2-3 and 1-4 both describe it, without changing the strip between checks. On an endpoint visit, align starts, let ends cross, and show how to measure the distance whichever end is farther right. Introduce the shared-versus-separate distinction in objects rather than only words.

Suggested visits: 15-25 minutes for linked rectangle sides; 15-30 for consistent/false cards; 15-30 for crossing ends and range design. A first route can stay with tile extrema. Keep the same hidden length while checking a row of cards; changing it between cards destroys the problem. Use one allowed witness per claim and save the card choices. The arithmetic-rich extensions are optional adult conversation.

Exact answers, hints, extensions

1. Enumerate A=1,2,3,4,5; the corresponding B values are 5,4,3,2,1. Areas are 5,8,9,8,5; the extrema are attained by 1-by-5 and 3-by-3. Rotations do not create new unlabeled shapes, but both A/B orders satisfy the problem. Neither 1 nor 25 occurs with the exact sum: 1 requires (1,1), sum 2, and 25 requires (5,5), sum 10. The honest tight interval is 5 to 9. Optional hint: can one more tile-width on one side preserve the sum without shortening the other? Extension: compare different fixed sums and use a cut-and-rearrange or $(A-B)^2$ argument for the most square-like rectangle; whole and real sides give different equality cases.

2. First row intersection is 5-6, including edges and intermediate values. Second row: the first card 1-4 may be false, with any length in 5-8 (e.g. 5); the last card 5-9 may be false, with any length in 3-4 (e.g. 3). The middle card cannot be the only false one because 1-4 and 5-9 have no shared value. Third row: only the first card 1-4 can be false; the two true cards intersect in 7-8 (e.g. 7), which indeed defeats 1-4. If the middle were false, 1-4 and 7-9 are incompatible; if the last were false, 1-4 and 6-8 are incompatible. Optional hint: keep two candidate true cards visible and ask for one endpoint they both permit, then check the discarded card really fails. Extension: invent a row with exactly two different possible false cards, or distinguish an inaccurate bound from an intentionally wide but true bound.

3. A=3-7, B=5-9: gap 0-6. Zero occurs at a common value such as 5; 6 at A=3,B=9. A can be longer (7,5), B can be longer (3,9), and ties are possible. A=2-4, B=6-8: gap 2-6. Minimum at 4,6, maximum at 2,8; B is always strictly longer and equality is impossible. A=3-5, B=5-7: gap 0-4. Zero only at 5,5; maximum at 3,7. B can be strictly longer, A never strictly longer, and equality is possible. Designed cards A=4-6,B=4-6 always give gap at most 2 and allow either order, for example 6,4 or 4,6. Many designs work; the task does not impose a widest-card optimum. Optional hint: try to make the ends coincide before looking only at opposing extremes. Extension: add a promise about their total and see which opposite extremes cease to be jointly possible.

Sources, newness, checks

Base Week 51 uses sums, shared-strip cancellation, disjoint longer/shorter gaps and exact-total lengths. Page 1 adds nonlinear area with linked sides; page 2 combines multiple guarantees for one object and diagnoses one false report; page 3 permits crossings/ties and asks for an absolute-distance guarantee and designed cards. Three investigations, one theme. Related return visit: Week 48 bonus, Problem 2, introduces combining area certificates; this hidden-length investigation adds diagnosis of an actually false report. Track that prior use rather than treating the shared certificate idea as a new theme.

Original extensions to the base interval arithmetic mathematics. NIST Digital Library of Mathematical Functions, §3.1(ii), equation (3.1.6), is identified in the base guide; these exact rectangle and false-card cases were independently derived. Week 1 encore and Math Circle by the Bay preface, printed pp. viii-x (PDF pp. 9-11), motivate return visits rather than specific tasks.

student/verify.py enumerates all five labeled rectangles, intersects all report sets, checks half-unit witnesses against the actually false card, and verifies all endpoint-gap extrema using half-unit enumerations and formulas. Continuous extrema follow from interval overlap and opposite-end arguments, not sampling alone. The equal designed cards are checked explicitly. No classroom or physical rehearsal is claimed.